APPENDIX A – MEDICALLY SIGNIFICANT ORGANISMS

PART I: THUMBNAIL SCETCHES.

This section provides brief sketches of medically important agents of infection. It has been adapted from Schaechter and colleagues excellent textbook "Mechanisms of Microbial Disease (3rd Edn.). It is intended to help introduce you to key features of these organisms that explain their disease associations, classification and epidemiology. You should read, understand and become familiar with all of this material. What you should learn remains defined in the learning outcomes set out at the beginning of each lecture.

BACTERIA

Staphylococci: Abscesses and Other Diseases

- Are pyogenic ("pus-forming") bacteria.
- Cause different diseases, including abscesses, food poisoning, and toxin-related diseases (toxic shock and scalded skin syndromes). If the organisms enter the bloodstream, they may cause osteomyelitis and endocarditis.
- Include *Staphylococcus aureus*, *Staphylococcus epidermis*, *Staphylococcus saprophyticus* (which almost exclusively causes urinary tract infections), and others.
- Possess a variety of virulence factors; *S. aureus* possesses cell-wall factors peptidoglycan, teichoic acid, and protein A), catalase, coagulase, and toxins. Some strains produce β-lactamase (an enzyme that breaks down penicillin).
- Infections are treated with antibiotics, although antibiotic resistance to penicillin and other antibiotics is a problem.

Streptococci

The streptococci:

- Can be classified according to haemolysis pattern, Lancefield group, or species
- Include Group A streptococci (*Streptococcus pyogenes*), which causes "strep throat," scarlet fever, and impetigo. Rheumatic fever and glomerulonephritis are nonsuppurative sequels of strep infection.
- Include Group B streptococci (*Streptococcus agalactiae*), the leading cause of neonatal sepsis and meningitis; Groups C and G streptococci, which cause infections similar to Group A infections; minute-colony streptococci, which cause abscesses; Group D streptococci, which include the enterococci and nonenterococci; and viridans streptococci, which cause subacute bacterial endocarditis and dental caries.

***Streptococcus pneumoniae,* the Pneumococcus**

- Is an α-haemolytic streptococcus;
- Is a frequent agent of pneumonias in adults as well as other localized infections, especially in children;
- Causes infections characterized by acute inflammation;
- Has a capsule that makes it resistant to phagocytosis;
- Occurs in many antigenic capsular types, some of which have been used to make a polyvalent vaccine.

Neisseriae: Gonococcus and Meningococcus

- Neisseriae are Gram-negative cocci

Neisseria gonorrhoeae (the gonococcus):

- CauseS gonorrhoea which can lead to pelvic inflammatory disease (PID) in women and epididymitis in men.
- Can be carried asymptomatically by both men and women.
- Exhibit phase variation and antigenic variation.
- Can cause disseminated infections (serum-resistant strains).
- Infections are treated with antibiotics, although many strains are resistant to penicillin.

Neisseria meningitidis (the meningococcus):

- Causes septicaemia and meningitis.
- Are heavily encapsulated and possess a haemolysin, which allows them to disseminate in the bloodstream, leading to disseminated intravascular coagulation (DIC).
- Can be prevented with vaccination with capsular polysaccharide, except for group B strains.

Bacteroides and Abscesses

Bacteroides:

- Are strict anaerobic Gram-negative rods that play a causative role in abscess formation.
- Are members of the normal microbiota of the gut and oral cavity.
- Includes *Bacteroides fragilis*, a bacterium that can withstand short exposure to oxygen; has a non-toxic lipopolysaccharide and a capsule; and produces several enzymes such as a neuraminidase, proteases and lipases that may play a role in its pathogenicity.

Abscesses formation in the periotoneal cavity:

- Begins when the colon wall is breached by a ruptured bowel, a ruptured appendix, or abdominal surgery and a large number of bacteria enter the peritoneal cavity.
- Is caused by a mixture of facultative and strict anaerobes (including *Escherichia coli*, *B. fragilis*, clostridia, and anaerobic streptococci); as infection proceeds, facultative anaerobes use up the available oxygen, and strict anaerobes become dominant.
- Is treated with surgical drainage and antibiotics; a combination of antibiotics or antibiotics sensitive to both facultative anaerobes and strict anaerobes is necessary.

Enteric Bacteria: "Secretory" (Watery) Diarrhoea

Enteric bacteria that cause secretory diarrhoea:

- Include members of Enterobacteriaceae and Virbionaceae.
- Are usually transmitted to humans in contaminated food and water.
- Face formidable defences in the gastrointestinal tract; they adhere to the epithelial cells of the GI tract via pili (fimbriae) or surface adhesin.
- May elaborate toxins (*Vibrio cholerae*, enterotoxigenic *Escherichia coli*) that reduce sodium absorption and increase chloride secretion of the intestinal cells, which leads to watery diarrhoea.
- Cause disease that can be treated with oral rehydration therapy.

Invasive and Tissue-damaging Enteric Bacterial Pathogens: Bloody Diarrhoea and Dysentery

The main invasive enteric pathogens are:

- *Shigella*, which cause dysentery and bloody diarrhoea. *Shigella* is closely related to enteroinvasive *Escherichia coli* (EIEC).
- Enterohemorrhagic *E. coli* (EHEC), which cause hemorrhagic colitis. The most common serotype is O157:H7.
- *Salmonella*, which can cause nontyphoidal gastroenteritis and typhoid fever; typhoid fever is a systemic infection of mononuclear phagocytes sometimes characterized by relapses and the carrier state caused by bacterial survival in the gallbladder.

***Pseudomonas aeruginosa*: Ubiquitous Pathogen**

Pseudomonas aeruginosa:

- Is a ubiquitous opportunistic pathogen found in soil and water.
- Has minimal nutritional requirements and survives in a variety of environments.
- Is a frequent cause of nosocomial (hospital-acquired) infections.
- Is an important pathogen in individuals with cystic fibrosis, burns, and diabetes.

***Bordetella pertussis* and Whooping Cough**

Bordetella pertussis infection:

- Causes whooping cough, or pertussis, a disease that causes characteristic paroxysmal cough in infants.
- Can be prevented with DPT, a vaccine containing β. *Pertussis* antigens, along with diphtheria and tetanus antigens.
- Is extremely contagious.
- Involves several virulence factors, including adhesins and toxins.

Clostridia

Clostridia:

- Are Gram-positive, anaerobic, spore-forming rods responsible for several unrelated diseases including pseudomembranous colitis, botulism, tetanus, gas gangrene, and cellulites.
- Often cause disease due to the production of toxins.
- Include pathogens such as *c. difficile*, which is associated with the use of antimicrobial drugs; botulism (caused by *C. botulinum*); tetanus (caused by *C. tetani*); and gas gangrene (caused by *c. perfringens*).

***Legionella*: Parasite of Cells**

Legionella:

- Are aerobic Gram-negative bacilli.
- Invade human macrophages.
- Have fastidious growth requirements and must be grown on special media.
- Cause waterborne infections such as Legionnaires' disease (legionellosis) and Pontiac fever.

Legionnaires' disease:

- Is caused by *Legionella pneumophila.*
- Affects only a minority of people exposed to the infectious agent. These people usually have risk factors such as smoking, advanced age, or pulmonary disease.
- Is cleared by a cell-mediated response.
- Causes damage in the lungs due to a vigorous host response to the infectious agent.
- Is difficult to diagnose, thus necessitating treatment based on suspicion alone.
- Is treated with antibiotics known to penetrate infected cells (erythromycin and tetracycline).
- Can be prevented by regularly checking institutional water systems for the presence of *L. pneumophila*.

***Helicobacter pylori:* Pathogenesis of a "Slow" Bacterial Infection**

Helicobacter pylori:

- Is a recently discovered bacterium that is the causative agent of most gastric and duodenal ulcers.
- Colonizes the mucus gel layer overlying the gastric mucosa because it produces urease, which converts urea to ammonia and increases the pH locally.
- Is extremely, common and probably infects up to half of the world's population.
- Causes clinical disease in only 5-10% of infected individuals.
- Is associated with adenocarcinoma of the stomach.
- Can be detected by histologic examination, culture, rapid urease testing, serologic testing, and urease breath tests.
- Is treated with a combination of antibiotic and acid-reducing drugs.

Mycobacteria: Tuberculosis and Leprosy

Tuberculosis:

- Is caused by *Mycobacterium tuberculosis*, a slow-growing, acid-fast obligate aerobe that invades host macrophages.
- Is transmitted by inhalation or ingestion.
- Can take two forms: primary tuberculosis, a usually mild disease; and secondary tuberculosis, a disease caused by reactivation of dormant organisms.
- Elicits a cellular immunity that usually halts the progression of the disease.
- Damage in the primary form includes the formation of granulomas followed by caseation; in immunocompetent patients, the lesions heal spontaneously; immunocompromised patients may proceed to military (disseminated) tuberculosis.
- Damage in the secondary form is caused by a delayed-type hypersensitivity reaction to reactivated organisms.
- Is diagnosed by a tuberculin skin test and microscopic and culture tests.
- Is treated with a combination of antibiotics (in order to thwart drug resistance) for 6 months.

Leprosy:

- Is caused by *Mycobacterium leprae*.
- Takes two forms: tuberculoid (a self-limiting disease) and lepromatous (the malignant form).

Syphilis: A Disease with a History

Syphilis:

- Is caused by the spirochete *Treponema pallidum*.
- Is characterized by three stages: primary syphilis, in which the organism is localized to a chancre, or skin lesion; secondary syphilis, in which the organism disseminates to distant sites; and tertiary syphilis, in which the immune response to the organisms leads to the destruction of tissue.
- Can be transmitted from mother to fetus (congenital syphilis).
- Is initially diagnosed by tests such as Venereal Disease Reference Laboratory Test (VDRL); positive results should be confirmed with more specific tests.
- Is readily treated with penicillin.

Lyme Diseases

Borrelia species:

- Are spiral-shaped spirochetes.
- Are transmitted to humans by tick bites.
- Cause Lyme disease and relapsing fever.
  Lyme disease is caused by *Borrelia burgdorferi*. A multiphased disease, it is treated with antibiotics.
  In Stage 1 (acute, local) the skin is affected at the site of the tick bite (erythema migrans lesion).
  In Stage 2 (disseminated) the central nervous system, the skin, and other systems are affected.
  In Stage 3 (chronic) the joints and the central nervous system are affected; the organisms are not usually found in the lesions, but their DNA can sometimes by detected.

Chlamydiae: Genital, Ocular, and Respiratory Pathogens

Chlamydiae:

- Include *Chlamydia trachomatis*, a sexually and vertically transmitted bacterium that causes genital tract infections; *Chlamydia pneumoniae*, which may be an almost universal pathogen of humans; and *Chlamydia psittaci*, an avian pathogens that can be transmitted to humans.
- Are obligate intracellular pathogens that act as energy parasites within the host cell.
- Cause damage by evoking both a cellular and humoral immune response.
- Can be treated with antibiotics that are able to enter host cells.

***Mycoplasma*: Curiosity and Pathogen**

Mycoplasma:

- Are the smallest organisms capable of growth on cell-free media.
- Lack a rigid cell wall.
- Usually require sterols for growth.
- Species that cause human disease include *Mycoplasma pneumoniae*, *Mycoplasma hominis*, and *Ureaplasma urealyticum.*

M. pneumoniae:

- Causes pneumonia limited to the respiratory mucosa that lines the airways.
- May cause mild haemolytic anaemia due to production of a cold hemagglutinin IgM antibody.
- Is usually diagnosed by it clinical features.
- Is treated with erythromycin or tetracycline.

M. hominis:

- Are transmitted through sexual contact as well as vertically.
- Cause pelvic inflammatory disease, chorioamnionitis, and postpartum fever.
- Have been isolated from the central nervous system of newborns.

U. urealyticum:

- Are transmitted through sexual contact as well as vertically.
- Cause urethritis.
- Are associated with chronic lung disease in very low-birth-weight premature infants.
- Cause pelvic inflammatory disease, chorioamnionitis, and postpartum fever.
- Have been isolated from the central nervous system of newborns.

VIRUSES

Viruses:

- Are obligate intracellular parasites that use the host cell's machinery to replicate.
- Contain a nucleic acid core surrounded by a protein capsid; the capsid of some viruses is surrounded by an envelope.
- Are absorbed into host cells by first binding to receptors on the host cell; the virus then enters the particles by receptor-mediated endocytosis or crossing the host cell membrane via coated pits or fusion with the host cell membrane (enveloped viruses).
- Have single- or double-stranded RNA or DNA, which, following the adsorption of the virus into the host cell, is then translated inside the host cell to make viral proteins.
- Assemble new virus particles within the host cells, which are then released during cell lysis (nonenveloped viruses) or by budding (enveloped viruses).

Viral infection:

- Can be lytic, latent, or chronic.
- Can be acquired from other humans or from the environment; direct contact routes include sexual contact and vertical transmission and environmental routes include respiratory (aerosols), gastrointestinal (fecal-oral contamination), and transcutaneous (inoculation).
- Is often characterized by an incubation period; the virus replicates within the host.
- Can spread via nerves to the nervous system and via the blood to many organs. Many viruses spread via multiple pathways.
- Is curtailed within the host primarily through cell-mediated immunity.
- Can be diagnosed through culturing in cell cultures, embryonated eggs, or animsla and immunocytochemical staining and identification of virus particles or antigens in tissue specimens.

Picornaviruses: Polio Virus, Other Enteroviruses, and the Rhinoviruses

Picornaviruses:

- Are a family of small RNA viruses.
- Include the enteroviruses, rhinoviruses, and the hepatitis A virus.

Enteroviruses:

- Derive their name from their natural habitat, the gastrointestinal tract.
- Include polioviruses, coxsackievirus, and the echoviruses (now known as numbered enteroviruses).

Polioviruses:

- Are positive-strand viruses that can cause either asymptomatic infection or flaccid paralysis.
- Replicate quickly in the intestine and can be shed in the feces weeks to months after infection.
- Can spread from the gastrointestinal tract to the central nervous system.
- Can be prevented with either a killed vaccine (Salk) or a live, attenuated vaccine (Sabin).

Rhinoviruses:

- Cause the common cold.
- Are distinguished into over 100 serotypes, which prevent development of an effective vaccine.
- Can be transmitted person-to-person through contamination or aerosols.
- Bind to specific receptors on respiratory epithelial cells.
- Probably cause damage by mechanisms other than viral-induced cytopathology.

Arthropod-borne Viruses

Arboviruses:

- Are viruses that are transmitted by an arthropod vector.
- Cause zoonoses; these diseases only accidentally affect humans.
- Are often named after the disease they cause or the place where they have been found (i.e. LaCrosse, Semliki Forest, Rift Valley fever).
- Cause encephalitis, dengue fever, and yellow fever.
- Are diagnosed by isolation of the virus or rise in antibody titer; diagnosis on clinical grounds is difficult.
- Cannot be treated, although vaccines against many arboviruses can prevent infection.

Paramyxoviruses: Measles, Mumps, Slow Viruses, and the Respiratory Syncytial Virus (RSV)

- Are enveloped RNA viruses
- Both were formerly common childhood infections, now rare and prevented by the MMR vaccine where this is accepted.
- Measles causes a rash, fever and occasionally respiratory complications
- Mumps is most often recognised through infection of the salivary glands and occasionally involves testicular or ovarian tissue, very rarely resulting in sterility in the former case.
- Both can have CNS involvement.
- RSV is a major cause of lower respiratory tract infections (bronchiolotis) in infants and increasingly recognised as an agent of LRT infections in adults, particularly the elderly.

Coronaviruses

- Are enveloped RNA viruses
- Show marked species and tissue specificity
- Include agents that cause the common cold and sever acute respiratory syndrome, SARS.

Rabies

- Is a zoonosis caused by the rabies virus, transmitted to humans by the bite of an infected wild or domesticated animal.
- In developing countries, is mainly a disease of dogs; in developed countries, of wild animals.
- Has a variable incubation period ranging from several weeks to about one year.
- Is spread from the site of the bite through nerves to the central and peripheral nervous systems; the virus then travels back to peripheral sites.
- Signs and symptoms include Negri bodies in neurons, difficulty swallowing, increased muscle tone, and hydrophobia.
- Is diagnosed with fluorescent antibody assay of skin biopsies early in the disease.
- Is prevented by vaccination of wild and domesticated animals.
- Is treated in humans with local wound treatment, passive administration of antibody, and vaccination.

Influenza and Its Virus

Influenza:

- Is caused by enveloped RNA viruses belonging to the Orthomyxovirus. Three strains (A, B, and C) infect humans.
- Epidemics and pandemics are the result of the virus' ability to undergo antigenic reassortments, leading to significant changes in viral antigens (notably, hemagglutinin and neuraminidase). The antigenically rearranged strains are usually of animal origin.
- Causes significant mortality in the elderly and individuals with underlying cardiopulmonary disease. Complications include bacterial superinfections and Reye's syndrome.
- Can be prevented by vaccination. A vaccine is prepared annually from strains most likely to emerge in the coming season.
- Can be treated or prevented with amantidine (A strains only), rimantidine, and other drugs.

Rotaviruses and Other Viral Agents of Gastroenteritis

Rotaviruses:

- Are wheel-shaped, double-stranded RNA viruses.
- Are the most common cause of gastroenteritis in children.
- Cause a disease with seasonal distribution in the U.S., with peak incidence in the winter months.
- Are transmitted by the fecal-oral route.
- Cause changes in the small intestine mucosa which lead to diarrhoea.
- Induce long-term immunity.

Other enteric viruses that cause gastroenteritis include:

- Norwalk virus, which is the leading cause of diarrhoea in adults.
- Two strains of adenovirus, which cause diarrhoea especially in children under the age of 2.
- Caliciviruses, which cause institutional outbreaks of diarrhoea.